Continuity & Differentiability

Mind map

Continuity
- $f(a^-) = f(a^+) = f(a)$
- Pen not lifted
- Only in the domain

Algebra of continuity
- Cont. $\pm$ cont. $=$ cont.
- Cont. $\pm$ discont. $=$ discont.
- Products: check

Interval
- $[a, b]$: right at a, left at b
- Checkpoints: pieces, [ ], sgn

Special functions
- Q/Q': where $g = h$
- $[f]$ by graph
- Built-in limit: $|x| = 1$

Discontinuity types
- Removable: missing / isolated
- Finite (jump), infinite, oscillatory
- Jump $= |\text{RHL} - \text{LHL}|$

Composite
- g discont. points
- $g(x) =$ discont. of f
- Or define piecewise

EVT and IMVT
- Max, min on $[a, b]$
- $f(a)f(b) < 0 \Rightarrow$ root
- $g =$ LHS $-$ RHS

Derivability
- LHD $=$ RHD $=$ finite
- Unique finite tangent
- Derivable $\Rightarrow$ continuous

Forms of $f'(a)$
- $\frac{f(a+h)-f(a)}{h}$, $\frac{f(x)-f(a)}{x-a}$
- $\frac{f(a+ph)-f(a+qh)}{h} \to (p-q)f'(a)$

LHD and RHD
- Finite unequal: corner
- Infinite: vertical tangent or break

Algebra of derivability
- Der. $\pm$ non-der. $=$ non-der.
- $g|f|$: need $g(a) = 0$

Counting non-derivable points
- Zeros of f in $|f|$
- Corners of max/min graphs
- $\sum |x - a_i|$: each a_i

Quick concept
- Continuous first
- $\lim_{x \to a^\pm} f'(x)$
- Not for oscillating f'

$x^p \sin \frac{1}{x}$
- Cont. $p > 0$
- Deriv. $p > 1$
- f' cont. $p > 2$

max/min over $[a, x]$
- Follow f to the peak, then flat
- Check junctions

Functional equations
- $f(0)$ from $x = y = 0$
- $f'(x)$ by first principle
- Or partial derivative